find :

- we can search a file based on some parameters.

  Parameters to use are

  1. Size
  2. Location
  3. Permission
  4. Owner
  5. Name

- Find provides real time search.

1. Find all files named passwd inside /etc directory

```
[student@node11 ~]$ find /etc -name passwd 2>/dev/null
/etc/passwd
/etc/pam.d/passwd
[student@node11 ~]$
```

2. Find all files named database (only files)

```
[student@node11 ~]$ find / -name database -type f 2>/dev/null
/var/lib/systemd/catalog/database
/home/student/database
[student@node11 ~]$
```

3. Find all folder named database (only directory)

```
[student@node11 ~]$
[student@node11 ~]$ find / -name database -type d 2>/dev/null
/tmp/database
[student@node11 ~]$
```

4. Find everything with the name database

```
[student@node11 ~]$ find / -name database 2>/dev/null
/tmp/database
/var/lib/systemd/catalog/database
/home/student/database
[student@node11 ~]$
```

5. Find files which is more than 10 MB in size

```
[student@node11 ~]$ find /usr -type f -size +10M 2>/dev/null
/usr/lib/firmware/mrvl/prestera/mvsw_prestera_fw-v2.0.img.xz
/usr/lib/firmware/mrvl/prestera/mvsw_prestera_fw-v3.0.img.xz
/usr/lib/firmware/mrvl/prestera/mvsw_prestera_fw-v4.0.img.xz
/usr/lib/firmware/mrvl/prestera/mvsw_prestera_fw-v4.1.img.xz
```

6. Find all files which are less than 10M in size

```
[student@node11 ~]$ find /etc -size -10M -type f 2>/dev/null
```

7. Find all files which are in the range of 5M to 10M

```
[student@node11 ~]$ find /usr -type f -size +5M -size +10M 2>/dev/null
/usr/lib/firmware/mrvl/prestera/mvsw_prestera_fw-v2.0.img.xz
/usr/lib/firmware/mrvl/prestera/mvsw_prestera_fw-v3.0.img.xz
/usr/lib/firmware/mrvl/prestera/mvsw_prestera_fw-v4.0.img.xz
```

8. Find all files has 777 permission

```
[root@node11 ~]# touch rohit.virus
[root@node11 ~]# echo date > rohit.virus
[root@node11 ~]#
[root@node11 ~]# chmod 777 rohit.virus
[root@node11 ~]#
[root@node11 ~]# ls
anaconda-ks.cfg  raja  rohit.virus
[root@node11 ~]#
[root@node11 ~]# find / -type f -perm 777 2>/dev/null
/root/rohit.virus
[root@node11 ~]#
```

9. Creating a file of size 100M

```
[root@node11 database]# dd if=/dev/zero of=newfile bs=1M count=100
100+0 records in
100+0 records out
104857600 bytes (105 MB, 100 MiB) copied, 0.405837 s, 258 MB/s
[root@node11 database]#
[root@node11 database]# ls -lh
total 101M
-rw-r--r-- 1 root root 100M Nov  7 10:50 newfile
-rw-r--r-- 1 root root  168 Aug 10  2021 passwd
[root@node11 database]#
```

10. Creating a file size 75M block size 5M

```
[root@node11 database]# dd if=/dev/zero of=newfile1 bs=5M count=15
15+0 records in
15+0 records out
78643200 bytes (79 MB, 75 MiB) copied, 0.602602 s, 131 MB/s
[root@node11 database]#
[root@node11 database]# ls -lh
total 176M
-rw-r--r-- 1 root root 100M Nov  7 10:50 newfile
-rw-r--r-- 1 root root  75M Nov  7 10:56 newfile1
-rw-r--r-- 1 root root  168 Aug 10  2021 passwd
[root@node11 database]#
```

11. Search all files inside /tmp which are bigger than 70M and copy them inside /tmp/bigfiles directory

```
[root@node11 ~]# find /tmp -type f -size +70M 2>/dev/null -exec cp -p {} /tmp/bigfiles/ \;
[root@node11 ~]# ls /tmp/bigfiles/
newfile  newfile1
[root@node11 ~]#
```

12. Preserving permission

```
[root@node11 ~]# find /tmp -name "file?????" -type f 2>/dev/null -exec cp -p {} /tmp/test4 \;
[root@node11 ~]# ls -l /tmp/test4
total 12
-rw-r--r-- 1 root    root    8777 Nov 29  2023 file.1.gz
-rw-r--r-- 1 student student    0 Nov  7 11:12 file1.mp3
-rw-r--r-- 1 root    root       0 Nov  7 11:17 file1.mp4
-rw-r--r-- 1 student student    0 Nov  7 11:12 file2.mp3
-rw-r--r-- 1 root    root       0 Nov  7 11:17 file2.mp4
-rw-r--r-- 1 student student    0 Nov  7 11:12 file3.mp3
-rw-r--r-- 1 root    root       0 Nov  7 11:17 file3.mp4
[root@node11 ~]#
```

13. Another find and exec

```
[root@node11 ~]# find /tmp -name "file?????" -type f 2>/dev/null -exec rm {} \;
[root@node11 ~]#
[root@node11 ~]# find /tmp -name "file?????"
[root@node11 ~]#
```

locate ⇒ reads one or more databases prepared by *updatedb*.

Database ⇒ /var/lib/mlocate/mlocate.db

locate passwd ⇒ Will search all files having exact name as passwd plus all files having the string passwd in the name or in the path of the file.

Differences between find and locate

1. find looks for an exact match but locate displays all files having an exact match, or the filename might be there as a string in the path also.
2. find is real time search but locate is not.
3. locate is faster in search than find

In minimal installation locate command is not present

To install locate command ⇒ # dnf install mlocate

To Start building the mlocate database ⇒ # updatedb

```
[student@node1 ~]$ locate file1
/home/student/file1
/home/student/brace/file1a.mp3
/home/student/brace/file1b.mp3
/home/student/brace/file1c.mp3
[student@node1 ~]$
```

grep : Global Regular Expression Print/Pattern

grep, egrep, fgrep - print lines matching a pattern

1. Searching a word in a file :

```
[student@node1 ~]$ grep devops /etc/passwd
devops:x:1001:1001::/home/devops:/bin/bash
[student@node1 ~]$
[student@node1 ~]$ grep rohit file1
[student@node1 ~]$ grep Rohit /home/student/file1
Rohit ki Khobor
[student@node1 ~]$
```

2. Output with line number

```
[student@node1 ~]$ grep -n Rohit /home/student/file1
1:Rohit ki Khobor
2:aaj Rohit ki khabi ??
3:Rohit toke sabir sir dakche
5:rohit name akjon Rohit Sharma Ache
[student@node1 ~]$
```

3. Displays all lines started with the pattern root

```
[student@node1 ~]$ grep -n '^Rohit' /home/student/file1
1:Rohitlization ki Khobor
3:Rohit toke sabir sir dakche
[student@node1 ~]$
```

4. Displays all lines ended with the pattern bash

```
[student@node1 ~]$ grep -n 'bash$' /etc/passwd
1:root:x:0:0:root:/root:/bin/bash
21:student:x:1000:1000::/home/student:/bin/bash
22:devops:x:1001:1001::/home/devops:/bin/bash
```

5. Displays all lines ended with the pattern Rohit

```
[student@node1 ~]$ grep -n 'Rohit$' /home/student/file1
2:aaj Rohitnitrate ki khabi Rohit
4:Ajj rohit biriyani khaoabe Rohit
[student@node1 ~]$
```

6. Displays nothing as there is no entry called ishan and STUDENT (Case Sensitive)

```
[student@node1 ~]$ grep -n ishan /home/student/file1
[student@node1 ~]$
[student@node1 ~]$ grep -n student /etc/passwd
21:student:x:1000:1000::/home/student:/bin/bash
[student@node1 ~]$
[student@node1 ~]$ grep -n STDUENT /etc/passwd
[student@node1 ~]$ grep -n STUDENT /etc/passwd
[student@node1 ~]$
```

7. Displays all lines having the pattern root irrespective of case

```
[student@node1 ~]$ grep -i ISHAN /home/student/file1
ishan bhalo chele
ISHAN lab Asst
[student@node1 ~]$
```

8. Displays all lines which are not ended with nologin

```
[student@node1 ~]$ grep -v 'nologin$' /etc/passwd
root:x:0:0:root:/root:/bin/bash
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
student:x:1000:1000::/home/student:/bin/bash
devops:x:1001:1001::/home/devops:/bin/bash
[student@node1 ~]$
```

9. Display Multiple line with multiple words

```
[student@node1 ~]$ grep -e Rohit -e ishan /home/student/file1
Rohitlization ki Khobor
aaj Rohitnitrate ki khabi Rohit
Rohit toke sabir sir dakche Rohit
Ajj rohit biriyani khaoabe Rohit
rohit name akjon Rohit Sharma Ache
ishan bhalo chele
```

10. Multiple word searching without case sensitivity

```
[student@node1 ~]$ grep -i -e Rohit -e ishan /home/student/file1
Rohitlization ki Khobor
aaj Rohitnitrate ki khabi Rohit
Rohit toke sabir sir dakche Rohit
Ajj rohit biriyani khaoabe Rohit
rohit name akjon Rohit Sharma Ache
ishan bhalo chele
ISHAN lab Asst
```

11. Multiple word searching + no case sensitivity + line number

```
[student@node1 ~]$ grep -n -i -e Rohit -e ishan /home/student/file1
1:Rohitlization ki Khobor
2:aaj Rohitnitrate ki khabi Rohit
3:Rohit toke sabir sir dakche Rohit
4:Ajj rohit biriyani khaoabe Rohit
5:rohit name akjon Rohit Sharma Ache
7:ishan bhalo chele
8:ISHAN lab Asst
```

12. Search a word starting with R ending with t

```
[student@node1 ~]$ grep -n R.*t /home/student/file1
1:Rohitlization ki Khobor
2:aaj Rohitnitrate ki khabi Rohit
3:Rohit toke sabir sir dakche Rohit
4:Ajj rohit biriyani khaoabe Rohit
5:rohit name akjon Rohit Sharma Ache
[student@node1 ~]$
[student@node1 ~]$
```